# Self-Compensating StrongARM Latch for High-Speed Energy-Efficient True Random Number Generators

Xinbo Huang, *Student Member, IEEE*, and Mahfuzul Islam, *Member, IEEE*

***Abstract*—This paper presents a self-compensating StrongARM latch topology tailored for high-speed and energy-efficient true random number generator (TRNG) design. Unlike conventional approaches that rely on auxiliary controllers to monitor the output and apply compensation, the proposed latch incorporates a coupling capacitor and internal feedback path, which autonomously mitigate offset voltages induced by transistor mismatch. This inherent self-compensation not only simplifies the overall architecture but also improves robustness against PVT variations. A TRNG prototype is realized by integrating four self-compensating StrongARM latches with a lightweight XOR-based post-processing stage, thereby ensuring high-quality randomness with minimal design complexity. The statistical properties of the generated bitstreams are verified through NIST SP 800-22 and NIST SP 800-90B test suites, confirming compliance with standard randomness and robustness requirements. Post-layout simulations in a 65-nm low-power CMOS process demonstrate that the proposed TRNG achieves a throughput of 2 Gbps while consuming only 0.107 pJ/bit, highlighting its suitability for energy-constrained security applications.**

***Index Terms*—TRNG, self-compensation, StrongARM latch, Markov chain, autocorrelation, Random number.**

## I. Introduction

With the explosive growth of the Internet of Things (IoT) and large-scale cloud platforms, high-quality random numbers have emerged as a fundamental resource for modern digital systems. Random numbers are the foundation of not only the cryptographic systems for key generation, data encryption, and mutual authentication, but also of probabilistic computing applications, such as stochastic neural network (SNN), quantum simulation, and combinational optimization [1].

An optimal random number generator (RNG) should effectively generate output bitstreams with a uniform probability distribution of 1 and 0, achieving the min entropy as close to unity as possible [2]. To achieve such characteristics, pseudo RNGs (PRNGs) are one potential approach. However, PRNGs built with deterministic algorithms suffer from seed leakage vulnerabilities and often exhibit excessive power consumption unsuitable for IoT cryptographic applications. Stochastic computing, which implements the PRNG, also suffers from low energy efficiency [3] and complex seed management [4]. Given these limitations, TRNGs harnessing natural physical entropy sources, such as thermal noise, telegraph noise, and oxide breakdown, have gained popularity due to their ease of integration into System-on-Chip (SoC) solutions at minimal cost [5]. Among these entropy sources, thermal noise is particularly advantageous because of its frequency-independent spectral characteristics.

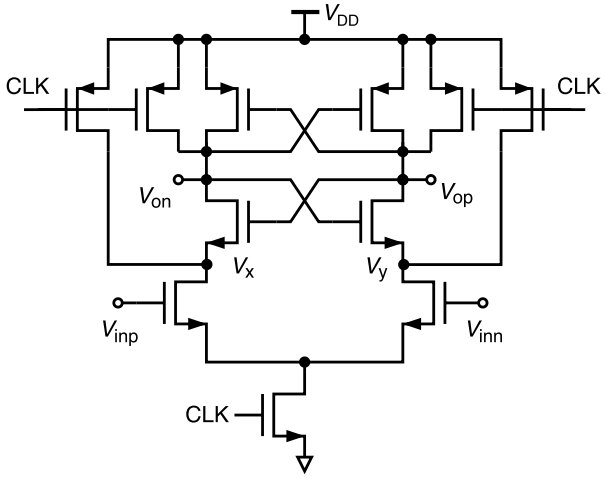



Fig. 1. Topology of the classical StrongARM latch.

Commonly used methods to extract the thermal noise include metastability of the latch [6], [7], [8], and jitter of the ring oscillator [9], [10], [11]. The primary design objective is to produce high-entropy random numbers with minimal post-processing, while maintaining low power consumption. The StrongARM latch, shown in Fig. 1, can serve as a circuit for both sampling and digitizing the thermal noise when both of its input terminals are shorted [12]. However, PVT (Process, Voltage, and Temperature) variations and transistor mismatch introduce offset voltage, resulting in a deterministic output. Therefore, continuous offset voltage compensation is indispensable for reliable TRNG realization. Existing

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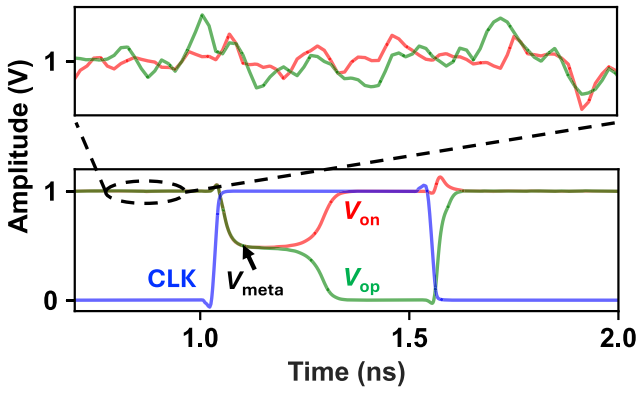



Fig. 2. Working principle of StrongARM latch-based TRNG with the input nodes connected to the supply voltage.

offset voltage compensation techniques suffer from either an energy-inefficient complex calibration procedure [6], [13] or a time-consuming auto-zeroing procedure [7], [8], [14].

In this work, we present a built-in offset voltage compensation mechanism that utilizes a coupling capacitor in conjunction with internal feedback connections, whereby charges of opposite polarity to the inherent offset voltage are autonomously accumulated. This approach eliminates the need for any auxiliary controller or the generation of additional analog or digital compensation signals. Benefiting from the simplicity of the self-compensating StrongARM latch structure, the proposed TRNG achieves both high throughput and superior energy efficiency. The implemented prototype, comprising four self-compensating StrongARM latches and three XOR gates, achieves a random bit generation rate of 2 Gbps with an energy consumption of only 0.107 pJ/bit.

The remainder of this paper is organized as follows. Section II reviews the StrongARM latch and existing compensation approaches. Section III introduces the proposed self-compensating StrongARM latch-based TRNG. Section IV develops an analytical model and presents a systematic design methodology. Section V presents simulation results and performance evaluation. Finally, Section VI concludes the paper.

## II. StrongARM Latch-Based TRNG

### A. Operating Principle

In an ideal situation where there is no mismatch, we can design a TRNG by simply connecting the input nodes of the StrongARM latch to the supply voltage. After the clock signal is asserted high, the cross-coupled inverter can converge to one of two stable states ($V_{op}$, $V_{on}$) = (1, 0) or (0, 1). Before settling into either of these states, however, the outputs $V_{op}$ and $V_{on}$ first enter a metastable state as shown in Fig. 2. At this metastable point, both nodes approach the logical threshold voltage, $V_{meta}$. The small voltage difference between $V_{op}$ and $V_{on}$, introduced by thermal noise, is then amplified by the positive feedback of the cross-coupled inverters. Once this difference grows sufficiently in either direction, the latch finally stabilizes to a valid logic state, producing either (1, 0) or (0, 1) at the outputs [15]. Because thermal noise has zero mean and flat power spectral density, the resulting bitstream exhibits equal probability of "1" and "0", as well as zero correlation between the consecutive bits.

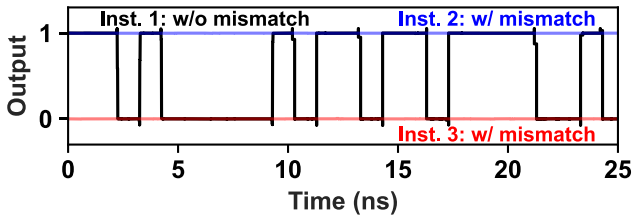



Fig. 3. Output waveforms of three StrongARM latch instances with input nodes connected to the supply voltage under thermal noise. The three instances represent three different mismatch scenarios.

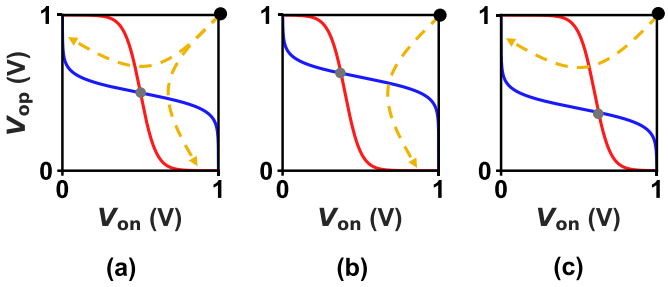



Fig. 4. Possible outputs of latch-based TRNG when (a) $V_{off} = 0$ (b) $V_{off} < 0$ (c) $V_{off} > 0$.

### B. Mismatch Effect

Despite its numerous advantages, including Gbps-level throughput, rail-to-rail output range, and robustness against power supply noise, StrongARM latch-based TRNG suffers significantly from the transistor mismatch introduced by PVT variations and transistor mismatch [6], [16]. Fig. 3 demonstrates the effect of transistor mismatch on the output bitstream using three different StrongARM latches, where the output waveforms under thermal noise are shown. The first instance corresponds to a case without a transistor mismatch. As the offset voltage is zero in this case, the output changes randomly between "1" or "0", which arises from the thermal noise-determined voltage fluctuations. The second and third instances include mismatches resulting in offset voltages with opposite polarity. For the latter two cases with nonzero offset voltages that are larger than the thermal noise-induced voltage fluctuations, the outputs are locked to either "1" or "0".

The detailed mechanism of how offset voltage determines the output is illustrated in Fig. 4. Fig. 4(a) depicts two transfer functions of the cross-coupled inverters without mismatch. The intersection point of the two transfer functions is defined as the metastable point. When no mismatch exists, two outputs $V_{op}$ and $V_{on}$ initially at state (1, 1), decrease symmetrically and simultaneously toward the metastable point. As the strength of pull-up and pull-down current matches exactly, $V_{op}$ and $V_{on}$ approach the metastable point with the same speed. While the two voltages approach the metastable point, noise will introduce a slight voltage difference between the two nodes. This difference will be amplified through the positive feedback of the two cross-coupled inverters. Once sufficiently amplified,

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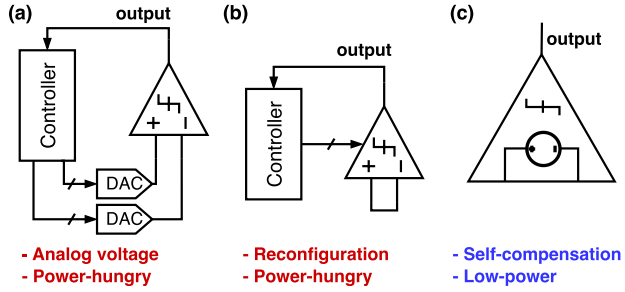



Fig. 5. Different compensation techniques. (a) DAC-based analog voltage generation. (b) Tuning transistor driving strength and timing skew. (c) Built-in technique (our proposal).

the balance is disrupted, driving the output nodes $V_{op}$ and $V_{on}$ towards either state (1, 0) or state (0, 1), which is determined by the polarity of the voltage difference.

However, transistor mismatch will reshape the transfer function by generating transconductance $g_m$ and threshold voltage $V_{th}$ differences between the symmetric transistors. Under mismatch, not only will the intersection point of the transfer function shift from the line $y = x$ line, but also the two output voltages $V_{op}$ and $V_{on}$ will no longer descend at the same speed. Consequently, two circumstances are generated as shown in Fig. 4(b) and Fig. 4(c). The final output thus depends on the polarity of the offset voltage instead of the thermal noise. It is worth noting that even a small offset-to-noise ratio of 0.05 results in a bias of 0.02, corresponding to a bit distribution of 52/48 between "1" and "0" [7]. More critically, the most typical offset-to-noise ratios range from 5 to 10, which means the output bitstreams of most StrongARM latches will always be 1s or 0s.

### C. Compensation Techniques

To compensate for the mismatch of the StrongARM latch-based TRNG, several techniques have been proposed. Fig. 5 illustrates two of these techniques along with our proposed one. Fig. 5(a) shows a technique where different input voltages are generated using DACs (Digital-to-Analog Converter) such that the offset voltage is canceled out [13]. A controller constantly monitors the output and tunes the analog input such that the probability of "1" approaches 0.5. Power consumption increases significantly in this case as the controller and the DACs need to operate continuously at high speed. Also, a large area is consumed by passive components and the controller circuits. Fig. 5(b) shows a technique where transistor driving strengths and clock timing are tuned digitally such that the output bitstream has equal probability of "1" and "0" [6]. Redundant transistors increase parasitic capacitance in each node, causing a significant power increase. The controller also becomes complex and, as such, consumes more power.

From the above discussion, it is evident that offset voltage compensation based on external feedback is power-hungry. Due to the resolution limitation, the compensation may not be accurate, which will introduce severe autocorrelation or bias to the output bitstream, necessitating extensive post-processing. This has been a fundamental obstacle in realizing energy-efficient TRNGs. Here, we propose and demonstrate that built-in self-compensation is possible with almost no additional power loss. Fig. 5(c) illustrates the concept of our proposed technique. The variable voltage source realized by a coupling capacitor generates a voltage with a polarity opposite to that of the offset voltage. We explain this efficient compensation technique in detail in the next section.

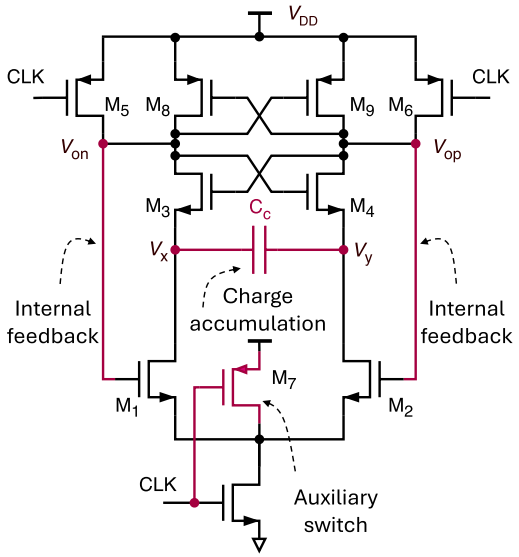



Fig. 6. The proposed self-compensating StrongARM latch with coupling capacitor and internal feedback paths.

## III. TRNG With Self-Compensating StrongARM Latch

### A. Self-Compensating StrongARM Latch

Fig. 6 illustrates the proposed self-compensating StrongARM latch [17]. Compared to the classical StrongARM latch, the proposed design introduces only two extra components: a coupling capacitor with capacitance $C_c$, and an auxiliary switch connected to the drain of the tail transistor. Simultaneously, the pre-charge switches at $V_x$ and $V_y$ are removed to prevent unintended reset of the accumulated compensation voltage. Moreover, two internal feedback paths are built by connecting the two input terminals to the output terminals accordingly. These internal feedback paths help to decide the polarity of the charge accumulation in each operating cycle of the StrongARM latch.

### B. Compensation Mechanism

First, the coupling capacitor strongly couples $V_x$ and $V_y$, thereby removing the offset voltage induced by the mismatch of the input differential pair. This coupling capacitor forces $V_x$ to increase or decrease at the same speed as $V_y$ when the input pair is on. It also stores a voltage difference $V_c$ between $V_y$ and $V_x$ when one of the input pair turns off. This phenomenon is demonstrated in Fig. 7, which shows how the output and the stored voltage $V_c$ change for two example instances with opposite offset polarity. Initially, the outputs remain constant because the $V_c$ is small and insufficient

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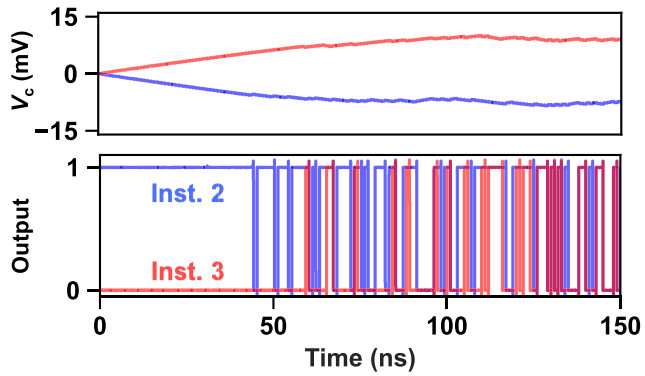



Fig. 7. Overall compensation effect with built-in charge accumulation.

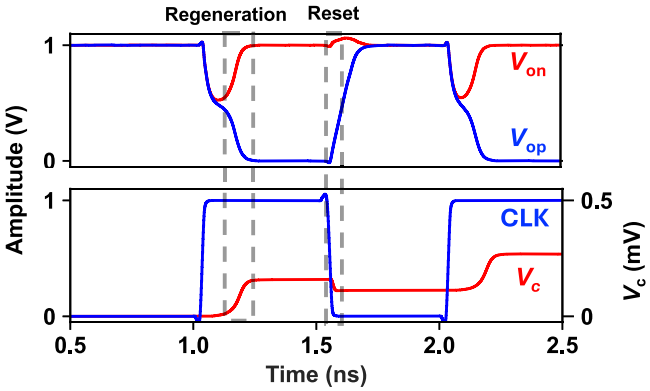



Fig. 8. Charge effect in one StrongARM latch operation cycle.

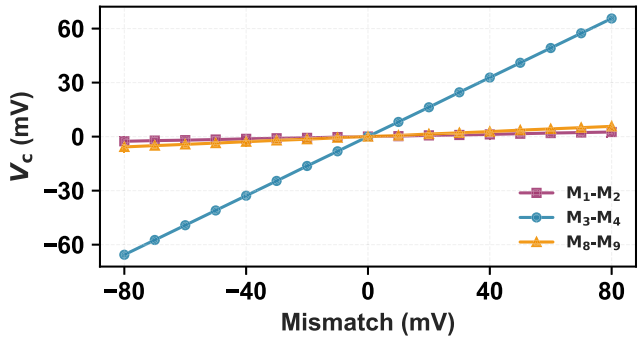



Fig. 9. Dependence of the average compensation voltage $V_c$ on the threshold mismatch of the input pair $M_1$-$M_2$, the nMOS latch pair $M_3$-$M_4$, and the pMOS latch pair $M_8$-$M_9$.

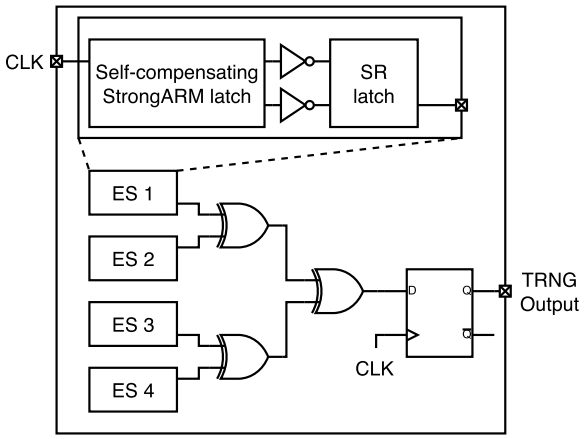



Fig. 10. Architecture of the proposed TRNG.

to cancel the offset voltage. However, during each cycle, a small amount of charge accumulates on the coupling capacitor, progressively reducing the offset. The output determines the polarity of this incremental voltage. As illustrated in Fig. 7, for instance #3, the output always keeps "0", and consequently, $V_c$ increases progressively until the probabilities of "1" and "0" become similar.

A detailed view of the compensation procedure within the initial few operation cycles for instance #3 is depicted in Fig. 8. In the regeneration phase of the StrongARM latch, after the output is determined, the residue charges in $V_{op}$ will flow into $V_y$. At the same time, the corresponding left input transistor turns off due to the decrease of $V_{op}$, allowing charges to accumulate on the coupling capacitor and increase $V_c$. Subsequently, in the reset phase, the extra auxiliary switch at the tail transistor drain node helps to charge $V_x$, preventing unwanted further $V_c$ increment. The magnitude of charge increased per cycle directly affects the output bitstream correlation, a relationship discussed in detail in the following section, motivating the addition of this auxiliary switch.

To quantitatively evaluate how $V_c$ responds to different levels of mismatch, Simulations are performed by intentionally injecting a controlled $V_{th}$ mismatch into different transistor pairs separately. Fig. 9 illustrates the dependence of the average compensation $V_c$ on the threshold voltage ($V_{th}$) mismatch of different MOSFET pairs in the StrongARM latch. It can be obtained that the mismatch of the nMOS latch pair $M_3$–$M_4$ requires the largest compensating voltage $V_c$, whereas only a small $V_c$ is needed for the mismatch of the input pair $M_1$–$M_2$ and the pMOS latch pair $M_8$–$M_9$. The dominant offset in a traditional StrongARM latch typically originates from the input differential pair, followed by the nMOS latch pair, while the contribution from the pMOS latch pair is generally negligible [12]. In our case, because the large coupling capacitor $C_c$ significantly suppresses the small-signal gain of the input pair, the mismatch contribution from $M_3$–$M_4$ becomes dominant, and the resulting offset voltage is the largest to be compensated by $V_c$.

### C. TRNG Prototype Design

As illustrated in Fig. 10, we design a TRNG prototype that consists of four entropy sources (ES), each implemented with a self-compensating StrongARM latch, two inverters, and one SR latch, followed by a 4-bit XOR stage and a final D flip-flop. Since the increment or decrement of $V_c$ is directly determined by the previous output bit, the mechanism inevitably introduces short-range autocorrelation into the bitstream. In addition, a small residual bias may remain due to incomplete offset cancellation. The XOR-based post-processing stage suppresses both effects, ensuring decorrelated outputs with full entropy.

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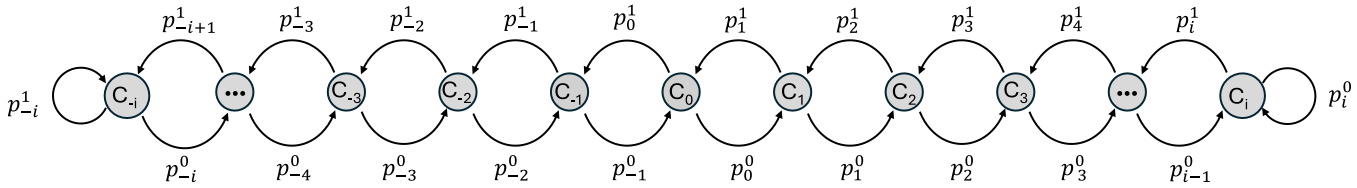



Fig. 11. Markov chain model of the proposed compensation mechanism.

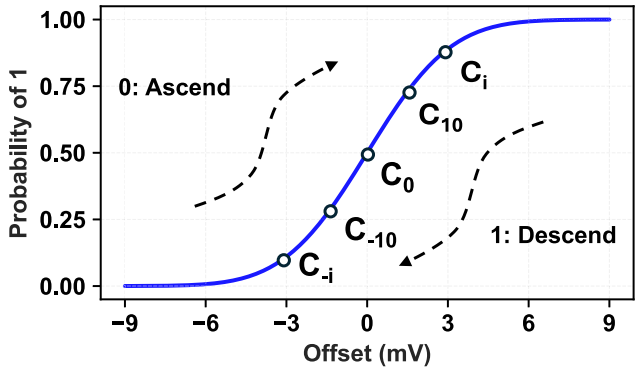



Fig. 12. Simulated cumulative distribution function (CDF) of the latch-based TRNG output probability, including the discrete compensation states ($C_i$) and the Ascend/Descend behavior of the proposed compensating mechanism.

## IV. Model Analysis and Circuit Design

All compensation methods inherently introduce some degree of autocorrelation in the TRNG output. Since lower correlation directly translates into reduced post-processing requirements, an important design consideration is to minimize the autocorrelation introduced by compensation.

In this section, the compensation behavior of the proposed self-compensating StrongARM latch is modeled using a Markov chain framework. This model captures the state transitions arising from the bit-dependent feedback mechanism and provides a quantitative means to evaluate the resulting autocorrelation. The obtained autocorrelation metrics then serve as a key guideline for sizing circuit parameters, thereby ensuring that the design achieves high randomness quality with minimal reliance on post-processing.

### A. Numerical Analysis

The proposed self-compensating StrongARM latch relies fundamentally on thermal noise, which exhibits a Gaussian distribution. Specifically, if an offset voltage between $V_x$ and $V_y$ is considered, a corresponding cumulative distribution function (CDF) can be obtained through observing the output probability. As shown in Fig. 12, a CDF curve is generated through circuit simulations assuming models for TSMC 65-nm technology at 1.2 V and 25 °C, resulting in a Gaussian distribution with $\mu = 0$ and $\sigma = 2.45$ mV.

Importantly, Fig. 12 not only shows the simulated CDF but also visualizes the compensation behavior: when the latch outputs "0", the operating point ascends along the CDF (increasing the probability of "1" in the next cycle), whereas output "1" causes a descent. Each $C_i$ corresponds to a discrete compensation state determined by a fixed offset tuning step. Ultimately, this dynamic feedback allows the latch-based TRNG to stabilize around the unbiased point $C_0$, ensuring an output probability converging to 0.5.

The behavior can be modeled using the Markov chain [16]. Ideally, the behavior of the random walk could be represented by a Markov chain with infinite states due to the wide and continuous Gaussian noise distribution. However, when the number of states increases to some extent, the state probability reaches 0, in which case, the Markov chain can be simplified to a model with limited states. Fig. 11 shows the Markov chain model of the proposed compensating method, where $P_i^{1/0}$ denotes the probability of 1/0 at state $C_i$. The transition mechanism of the Markov chain is given by:

$$S_i \cdot P_i^1 = S_{i-1} \cdot P_{i-1}^0, \tag{1}$$

$$\sum_{i=-n}^{n} S_i = 1. \tag{2}$$

Here, $S_i$ denotes the occurrence probability of the state $C_i$. These state-dependent probabilities $P_i^{0/1}$ could be directly obtained from the CDF curve. With Eqs. (1) and (2), state probabilities can also be easily calculated.

Using the calculated probabilities, the autocorrelation characteristics of the proposed compensating mechanism can be predicted. While the random bits contain only 1s and 0s, the autocorrelation factor $\phi_k$ can be expressed in a simplified form as [18]:

$$\phi_k = \frac{\mathrm{Cov}(X_t, X_{t+k})}{\sigma_X^2} = \frac{\sum_{t=1}^{n-k}(X_t-\mu)(X_{t+k}-\mu)}{\sum_{t=1}^{n}(X_t-\mu)^2} = \frac{P_{11}(k)-\mu^2}{\mu\cdot(1-\mu)}, \tag{3}$$

where $\mu$ denotes the mean output probability. Here, $P_{11}(k)$ represents the probability of observing consecutive "1"s separated by lag $k$ and can be represented as:

$$P_{11}(k) = \sum_{i=-16}^{16}\sum_{j=-16}^{16} S_i P_i^1 T_{i-1}^j (k-1) P_j^1, \tag{4}$$

$$\mu = \sum_{i=-16}^{16} S_i P_i^1, \tag{5}$$

where $T_{i-1}^j(k-1)$ represents the transition probability from state $C_{i-1}$ to state $C_j$ after $k-1$ cycles.

Taking the tuning step of 150 $\mu$V as an example, the maximum autocorrelation occurs at lag 1 with $\phi_1 = -0.024$ and gradually decays to approximately zero by lag 50. This

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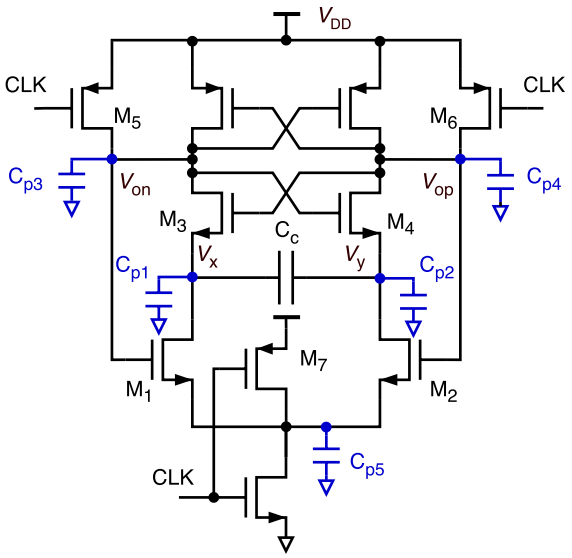



Fig. 13. Proposed self-compensating StrongARM latch including parasitic capacitances.

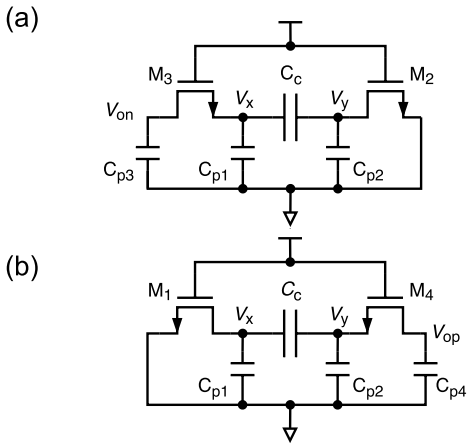



Fig. 14. Equivalent circuit in regeneration phase when the final result would be (a) $V_{op} = 0$, (b) $V_{op} = 1$.

level of correlation is sufficiently small to be eliminated by standard XOR-based post-processing.

### B. Design Procedure

To reduce short-range autocorrelation in the bitstream, we minimize the offset-tuning step, which equals the $\Delta V_c$ across the coupling capacitor $C_c$ in one operating cycle. As shown in Fig. 8, $\Delta V_c$ is set primarily during regeneration, with only a minor contribution during reset. Equivalent-circuit models for both phases are therefore employed to find the most appropriate circuit sizing.

The first increase happens when the positive feedback of the cross-coupled inverter has determined the output. $V_{op}$ and $V_{on}$ will be regenerated to 1 and 0 seperately. As illustated in Fig. 13, $C_{p1}$ and $C_{p2}$ correspond to the parasitic capacitances at the internal latch nodes $V_x$ and $V_y$, while $C_{p3}$ and $C_{p4}$ are associated with two output nodes $V_{on}$ and $V_{op}$, respectively. Fig. 14(a) and Fig. 14(b) show two different cases during

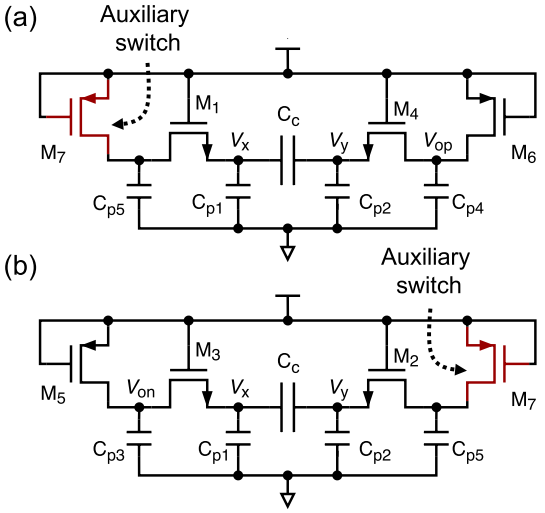



Fig. 15. Equivalent circuit in reset phase when (a) $V_{op} = 0$, (b) $V_{op} = 1$.

the regeneration phase. Take the case when $V_{op}$ will finally turn to 0 as an example, as shown in Fig. 6, $V_{op}$ and $V_{on}$ firstly reach the threshold voltage of the inverter at the same speed. So, the input differential pair also have the same drive strength, causing no difference between the two nodes of the coupling capacitor. Until the time when the thermal noise-induced difference is amplified to a large degree, when $V_{op}$ begins to drop, and $V_{on}$ begins to increase. When $V_{op}$ reaches the threshold voltage of $M_2$, the overall equivalent circuit is shown in Fig. 14(a). The residual electric charge in $C_{p4}$ will be discharged through the coupling capacitor $C_c$ to the ground. Through Kirchhoff's Current Law in $V_x$ and $V_y$, we get:

$$C_c \frac{d(V_y - V_x)}{dt} = \frac{V_x}{r_{o1}} + C_{p1} \frac{dV_x}{dt}, \tag{6}$$

$$C_c \frac{d(V_y - V_x)}{dt} = \frac{V_{op} - V_y}{r_{o4}} - C_{p2} \frac{dV_y}{dt}. \tag{7}$$

Here, $\frac{V_x}{r_{o1}}$ represents the current flow through the $M_1$ and $\frac{V_{op}-V_y}{r_{o4}}$ represents the current flow through the $M_4$. By the principle of superposition, we first integrate both Eq. (6) and Eq. (7), and then sum the results. Substituting $V_c$ for $V_y - V_x$ yields:

$$\Delta V_c = \frac{2Q + 2C_{p1}\Delta V_x}{C_c}. \tag{8}$$

Here, $Q$ represents all the electric charges stored in $C_{p4}$, which could be approximated through $C_{p4}V_{th2}$. Because after $M_2$ is totally off, all the charges begin to flow to the coupling capacitor $C_c$.

Similarly, when $V_{op}$ finally turn to 1, we obtain:

$$\Delta V_c = -\frac{2Q + 2C_{p2}\Delta V_y}{C_c}. \tag{9}$$

Here, $Q$ is approximated through $C_{p3}V_{th1}$. To achieve the minimum $\Delta V_c$, not only is a large coupling capacitor needed, but also the parasitic capacitances $C_{p1} \sim C_{p4}$ should be minimized.

Next, the equivalent circuit of the minor $\Delta V_c$ in the reset phase is shown in Fig. 15(a) and Fig. 15(b). $C_{p5}$ represents

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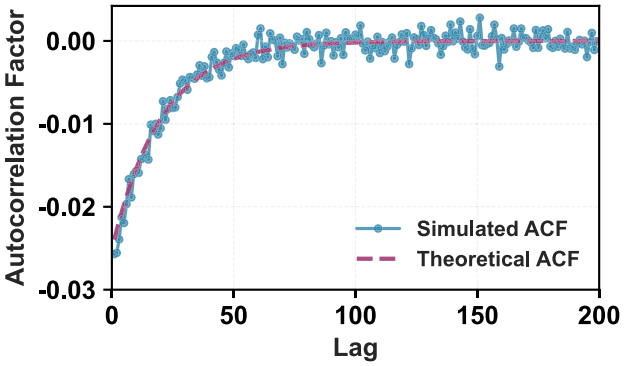



Fig. 16. Autocorrelation test with 1 million bits simulated from the proposed self-compensating StrongARM latch.

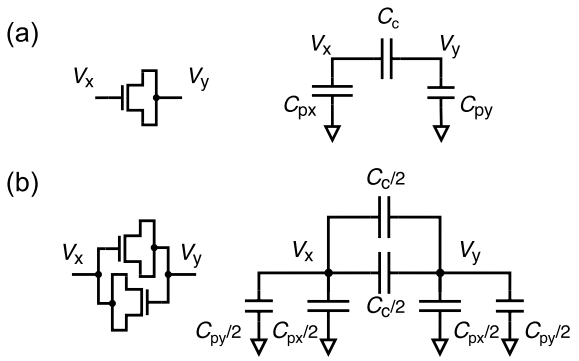



Fig. 17. (a) Single nMOSCAP and its equivalent capacitance model. (b) Two inversely connected nMOSCAPs and their equivalent capacitance model.

all the parasitic capacitance at the drain of the auxiliary switch, which is implemented to counteract the charge to the coupling capacitor from the pre-charge switch, preventing the $\Delta V_c$ increase or decrease again in the same operating cycle. Through the appropriate sizing of the auxiliary switch, an almost-balanced charge to $C_c$ could be achieved, resulting in a negligible $\Delta V_c$ in the reset phase.

With Eq. (8) and Eq. (9), the required value of the coupling capacitor can be estimated. Assuming typical device sizing in the StrongARM latch, the parasitic capacitances $C_{p1}$ and $C_{p2}$ are approximately 1.5 fF each, while $C_{p3}$ and $C_{p4}$ are about 0.7 fF. The threshold voltage $V_{th}$ and the voltage differences $\Delta V_x/\Delta V_y$ are taken as 0.4 V and 0.1 V, respectively. Based on these values, achieving a $\Delta V_c$ of 150 $\mu$V requires a coupling capacitor of about 8.9 pF.

### C. Verification

Using the calculated coupling capacitor and typical device sizing, a self-compensating StrongARM latch was implemented and simulated to generate one million output bits for autocorrelation analysis. Fig. 16 compares the theoretical ACF derived from Eq. (3) with the simulated ACF. The two results show excellent agreement: the maximum correlation occurs at lag 1 and gradually decays to zero by around lag 50. This consistency validates both the analytical model and the effectiveness of the proposed design.

Intuitively, the short-lag autocorrelation originates from the compensation mechanism itself. Each state is slightly influenced by the previous one via $V_c$, making adjacent bits weakly correlated. As the lag grows, this dependency diminishes and the correlation vanishes. A smaller offset-tuning step causes less change in $V_c$ between consecutive states, thereby reducing the statistical dependence and resulting in lower autocorrelation. This intuitive relationship is consistent with Eqs. (3) and (5), where a smaller step between adjacent states leads to lower autocorrelation, and the correlation naturally decreases as the lag increases.

### D. Layout Design

Inappropriate layout implementation can severely degrade the compensation effect. From Eqs. (8) and (9), it is clear

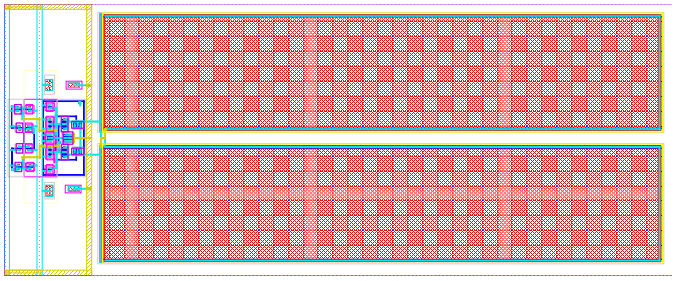


Fig. 18. Layout of one entropy source.

that any mismatch between $C_{p1}$ and $C_{p2}$ results in unequal ascending and descending, ultimately leading to biased outputs. To maintain balanced compensation while minimizing chip area, we adopt the capacitor with the highest density: the nMOSCAP [19]. When implementing the coupling capacitor with a single nMOSCAP, the equivalent circuit is shown in Fig. 17(a). In this structure, the parasitic capacitance to ground $C_{px}$ at the gate terminal dominates over that at the drain–source side $C_{py}$, with $C_{px}$ scaling primarily with gate width. Therefore, two nMOSCAPs, each with a maximum length of 50 $\mu$m and a width of 10 $\mu$m, are selected to achieve balance, as illustrated in Fig. 17(b). This configuration yields a total effective coupling capacitance of approximately 8.6 pF, corresponding to a compensation step $\Delta V_c$ of approximately 150 $\mu$V. Fig. 18 shows a layout example of the self-compensating StrongARM latch, which consists of a classical StrongARM latch and two nMOSCAPs. The entropy source occupies an area of 1200 $\mu m^2$.

## V. Simulation Results

The self-comepsating StrongARM latch-based TRNG is designed and simulated using TSMC 65-nm low-power CMOS technology. The mismatch compensating effect and the autocorrelation are first checked. Then the NIST SP800-22 [20] and NIST SP800-90B [21] are implemented to check the robustness and effectiveness of the proposed self-compensating StrongARM latch. The Shannon entropy and the min entropy under different temperatures and supply attack frequencies are then examined. Finally, a comparison with other state-of-the-art solutions was summarized.

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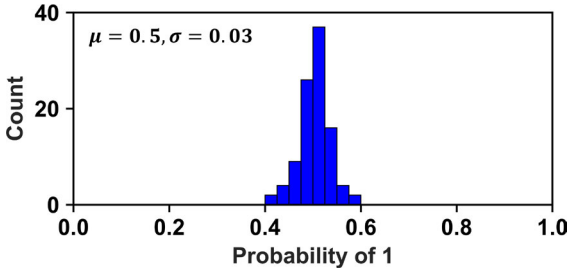



Fig. 19. Histogram of output probability with 100 Monte Carlo trials and 2000 bits each.

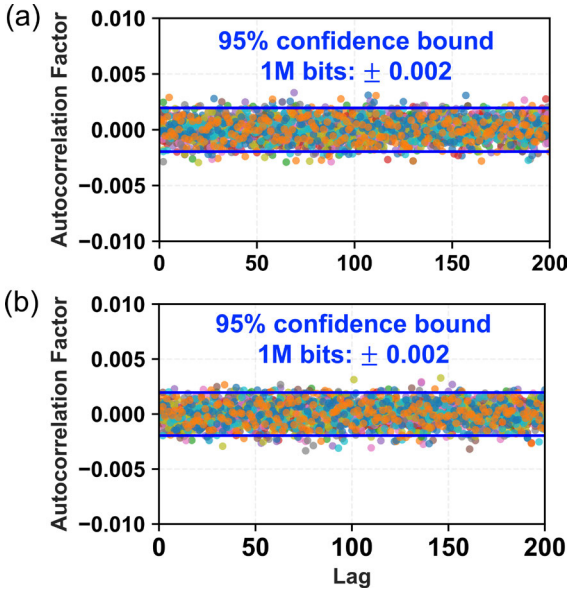



Fig. 20. Simulated autocorrelation results of the proposed TRNG: (a) with normal operating conditions (12 cases); (b) with 1 MHz, 200 $mV_{pp}$ supply voltage attacks (12 cases).

### A. Bias and Autocorrelation Analysis

To validate the efficiency of mismatch compensation, 100 times Monte Carlo simulations are conducted. Each collects 2000 bits to determine the distribution of output probabilities. Results in Fig. 19 indicate the self-compensating StrongARM latch achieves a mean output probability $\mu = 0.50$ with a standard deviation $\sigma = 0.03$, ensuring high Shannon entropy after the XOR-based post-processing. The final output obtained from 12 Monte Carlo simulations at a supply voltage of 1.2 V and a clock frequency of 2 GHz achieves the highest Shannon entropy of 0.99999999997 and an average min entropy of 0.995.

As illustrated in Fig. 10, ACF tests were performed on the outputs after XOR gate-based post-processing, with each test using 1 million bits generated by the TRNG operating at 2 GHz. Fig. 20(a) presents the results of 12 Monte Carlo simulations under normal operating conditions without power supply attack, while Fig. 20(b) shows the results under a power supply attack of 200 $mV_{pp}$ at 1 MHz. In both cases, the ACF distributions remain flat, with most autocorrelation factors falling within the 95% confidence bound, demonstrating that no significant autocorrelation exists in the final output bitstream.

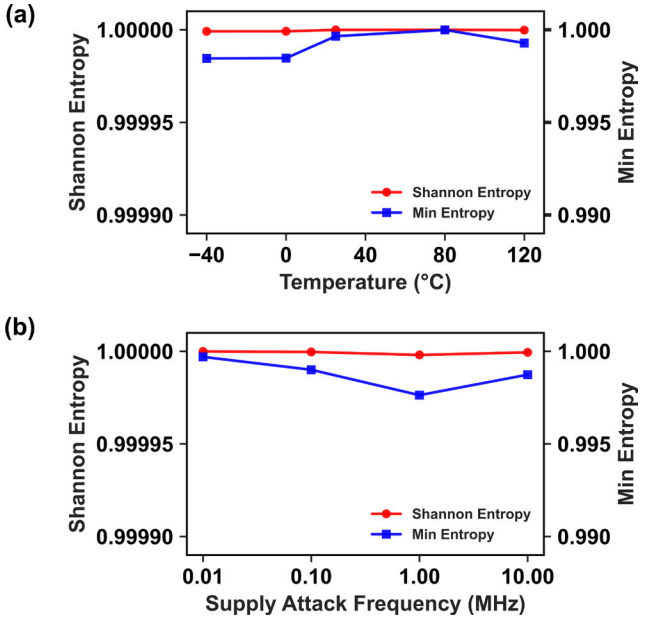



Fig. 21. Shannon entropy and min entropy with (a) temperature variation, (b) power attack.

### B. Randomness Verification

The NIST SP 800-22 test is implemented to verify the randomness. For each subtest, the passing criterion requires a P-value greater than 0.01, with an acceptable pass rate of 0.904 when applied to 12 bitstreams. Table II summarizes the results of twelve 1-million-bit sequences generated from independent Monte Carlo simulations at a supply voltage of 1.2 V and a temperature of 25 ° C. To verify the supply attack tolerance, 12 additional 1-million bitstreams are obtained with a power supply fluctuation of 1 MHz, which has a peak-to-peak voltage of 200 mV. In all cases, the minimum pass rate across subtests was 11/12, exceeding the required threshold of 0.904. It should be noted that not all generated bitstreams are applicable to the Random Excursion (Variant) test. Therefore, only eight results are reported. This restriction stems from the test specification itself and does not reflect any deficiency in randomness quality.

The generated bitstreams are further validated through the NIST SP800-90B Non-IID test, which provides various min entropy estimation methods by evaluating different potential patterns in the data. As summarized in Table III, the estimated min entropy remains stable across ten different estimation methods. To assess robustness, entropy was also analyzed under varying operating conditions. Fig. 21(a) presents the measured Shannon entropy and min entropy across temperatures ranging from −40 ° C to 120 ° C, confirming stable operation over a wide temperature range. Fig. 21(b) presents the results under a supply attack with a 200 $mV_{pp}$ supply attack with frequencies from 0.01 MHz to 10 MHz, further demonstrating the resilience of the proposed TRNG against harsh conditions. The slight drop observed at 1 MHz can be attributed to statistical variations arising from the finite sample size (1 Mbits) used in the min entropy estimation, rather than any degradation of the entropy source itself. As verified in Table III, the observed variation ($\sigma = 0.001$) aligns with that expected from the same estimation performed without

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TABLE I
COMPARISON WITH STATE-OF-THE-ART SOLUTIONS

| | JSSC'2016 [1] a) | ISSCC'2017 [10] a) | SSCL'2018 [13] a) | JSSC'2019 [5] a) | JSSC'2022 [8] a) | VLSI'2025 [7] a) | SSCL'2025 [14] a) | This Work b) |
|---|---|---|---|---|---|---|---|---|
| Entropy Source | Metastability | Jitter | Metastability | Metastability | Metastability | Metastability | Metastability | **Metastability** |
| Technology [nm] | 14 | 65 | 65 | 14 | 130 | 4 | 28 | **65** |
| Throughput [Mbps] | 162.5 | 9.9 | 86.0 | 1480 | 2.39 | 10 | 162.5 | **2000** |
| Supply [V] | 0.75 | 1.08-1.2 | 0.53 - 1.0 | 0.55 - 0.75 | 0.3 - 1.0 | 0.75 | 0.3-0.9 | **1.2** |
| Power [μW] | 1500 | 289 | 523 | 3700 | 10 | 17 | 22.75 | **214** |
| Energy [pJ/bit] | 9.23 | 35.24 | 6.08 | 2.5 | 2.5 | 1.7 | 0.140 | **0.107** |
| Post-processing | XOR based SFR, BIW Extractor | XOR | Markov Chain, LFSR | Hierarchical VN (off-chip) | XOR, VN8W | XOR | VN4 | **XOR** |

a) Measurement
b) Post-layout simulation

TABLE II
NIST SP800-22 TEST RESULTS

| NIST SP800-22 | Without supply attack | | With supply attack 1 MHz, 200 $mV_{pp}$ | |
|---|---|---|---|---|
| | Ave. P-value | Pass rate | Ave. P-value | Pass rate |
| Frequency | 0.43 | 11/12 | 0.51 | 12/12 |
| Block Frequency | 0.61 | 12/12 | 0.62 | 12/12 |
| Cumulative Sums (Forward) | 0.33 | 11/12 | 0.54 | 12/12 |
| Cumulative Sums (Reverse) | 0.39 | 12/12 | 0.553 | 12/12 |
| Runs | 0.53 | 12/12 | 0.60 | 12/12 |
| Longest Runs | 0.48 | 12/12 | 0.69 | 12/12 |
| Rank | 0.44 | 12/12 | 0.57 | 12/12 |
| FFT | 0.46 | 12/12 | 0.38 | 12/12 |
| Non-overlapping | 0.42 | 12/12 | 0.38 | 12/12 |
| Overlapping | 0.48 | 11/12 | 0.48 | 12/12 |
| Universal | 0.43 | 12/12 | 0.48 | 12/12 |
| Approximate Entropy | 0.49 | 12/12 | 0.55 | 12/12 |
| Random Excursions | 0.45 | 8/8 | 0.49 | 8/8 |
| Random Excursions (Variant) | 0.36 | 8/8 | 0.46 | 8/8 |
| Serial | 0.51 | 12/12 | 0.52 | 12/12 |
| Linear Complexity | 0.52 | 12/12 | 0.39 | 12/12 |

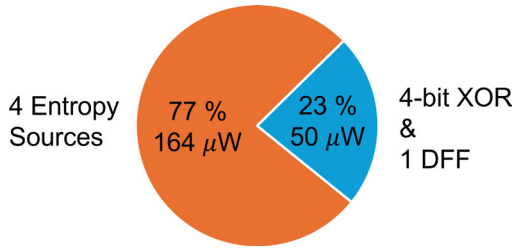



Fig. 22. Breakdown of power consumption.

$V_{DD}$ variation, indicating that the TRNG maintains consistent randomness and robustness with supply attack.

### C. Energy Consumption and Throughput

The proposed solution achieves an ultra-low power consumption of 214 $\mu$W with a throughput as high as 2 Gbps at 1.2 V, translating to an energy consumption of 0.107 pJ/bit. The energy consumed by the post-processing and DFF takes around 23 % of the total power, as shown in Fig. 22. When operating at 1.0 V, it further reduces the power consumption to 75 $\mu$W, offering a throughput of 1 Gbps with reduced energy consumption of 0.075 pJ/bit.

TABLE III
NIST SP800-90B TEST RESULTS

| NIST SP800-90B | Without supply attack | | With supply attack 1 MHz, 200 $mV_{pp}$ | |
|---|---|---|---|---|
| | Avg | Std | Avg | Std |
| **Non-IID Min-entropy estimation** | | | | |
| MCV | 0.995 | 0.001 | 0.995 | 0.001 |
| Collison | 0.903 | 0.016 | 0.909 | 0.013 |
| Markov | 0.998 | 0.001 | 0.998 | 0.001 |
| Compression | 0.841 | 0.018 | 0.855 | 0.022 |
| T-Tuple | 0.931 | 0.011 | 0.935 | 0.011 |
| LRS | 0.980 | 0.023 | 0.969 | 0.040 |
| MultiMCW | 0.996 | 0.001 | 0.996 | 0.001 |
| Lag | 0.996 | 0.002 | 0.995 | 0.004 |
| MultiMMC | 0.986 | 0.034 | 0.996 | 0.001 |
| LZ78Y | 0.986 | 0.034 | 0.997 | 0.001 |

### D. Comparison

Table I summarizes the overall performance of the proposed TRNG and other state-of-the-art solutions. Compared with prior works, where [5] reports the highest throughput and [7] achieves the lowest energy consumption, the proposed TRNG prototype operates stably with a supply voltage of 1.2 V, achieves the even higher throughput of 2 Gbps without any extra boost in the post-processing circuit, and has the lowest energy consumption at 0.107 pJ/bit.

## VI. CONCLUSION

This paper presented a built-in self-compensating StrongARM latch topology that employs a coupling capacitor to

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cancel offset voltage, thereby eliminating bias effectively. By integrating four such latches with a lightweight XOR-based post-processing circuit, which suffices to decorrelate and debias the raw output, the proposed TRNG achieves high-quality random number generation at a throughput of 2 Gbps with an energy cost of only 0.107 pJ/bit. Post-layout simulations using TSMC 65-nm low-power CMOS transistor models confirm the effectiveness of the self-compensating StrongARM latch and validate both the randomness and robustness of the generated bitstreams. Owing to its simplicity, scalability, and energy efficiency, the proposed architecture is highly suitable for hardware security modules, Internet-of-Things (IoT) devices, and other energy-constrained applications. Future work will focus on silicon implementation and the extension of the proposed topology toward large-scale TRNG arrays and hybrid TRNG–PUF architectures.

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